

ABHIRAMI AJITH KUMAR BINDU

Kerala, India | +91 7025413711 | abhiramiab242@gmail.com | [LinkedIn](#)

RESEARCH PROFILE

MRes Cancer Biology candidate at Imperial College London with research experience in neuroblastoma biology, DNA damage response, RNA splicing and cancer therapeutics. Currently investigating the impact of CDK12/13 inhibition and RBM39 degradation on transcriptional stress, DNA repair and immune activation in neuroblastoma.

EDUCATION

Imperial College London, United Kingdom

MRes Cancer Biology | Sept 2025 – Sept 2026 (Expected)

Sree Chitra Thirunal College of Engineering, India

BTech (Honours) Biotechnology & Biochemical Engineering | 2021

CGPA: 9.2

RESEARCH EXPERIENCE

Master's Research Project, Imperial College London, United Kingdom

Dec 2025 – Present

Investigating the impact of CDK12/13 inhibition and RBM39 degradation on RNA splicing, DNA damage repair and innate immune activation in neuroblastoma

- Studying transcriptional stress induced by CDK12/13 inhibitors and RBM39 degraders in MYCN-amplified and non-MYCN-amplified neuroblastoma cell lines.
- Evaluating DNA damage response pathways through RT-qPCR analysis of DNA repair genes, γ H2AX and Geminin immunofluorescence imaging, and RAD51 foci formation assays.
- Investigating RNA splicing alterations through RT-PCR-based exon skipping analysis and RT-qPCR quantification of intron retention events.
- Assessing innate immune activation through Western blotting, RT-qPCR and ELISA.
- CRISPR-Cas9-mediated gene knockdown
- Performing apoptosis quantification using Caspase-Glo 3/7 assays following targeted therapeutic interventions.
- Conducting homologous recombination (HR) reporter assays and analysing DNA repair efficiency using flow cytometry.
- Analysing and validating RNA-seq datasets to investigate transcriptional and splicing alterations induced by CDK12/13 inhibition and RBM39 degradation.
- Utilising confocal microscopy, InCell imaging systems for quantitative cellular analysis.

Student Researcher Imperial College London, United Kingdom

26/11/2025 – 05/12/2025

8 days lab practicals on the study of Lapatinib on HER2-positive patient-derived cell lines.

Hands on training on:

- 3D spheroids and 2D cell culture on cancer patient cell lines
- Crystal violet assay, Lapatinib drug treatment
- RT-qPCR, Western blotting and immunofluorescence staining

Research Intern Regional Cancer Centre – Trivandrum, Kerala, India

01/01/2025 – 31/01/2025

Exposure to clinical cancer cytogenetics and molecular diagnostics including:

- Chromosome separation from patient's bone marrow sample and Q banding
- Fluorescence Insitu Hybridization Technique to detect Acute Myeloid Leukemia and Chronic Myeloid Leukemia
- Analyzing the anticancer activity of drug verteporfin on SK-BR 3 cell line by MTT assay, Wound healing assay, Trypan blue staining and AO/EtBr staining
- Hands on training on qPCR

Research Intern Armats Biotek Training And Research Institute - Chennai, India

21/10/2024 – 21/11/2024

- Hands-on training on various biochemical assays, microbial assays, DNA extraction and purification.

Project intern Bioroot Exploration India Pvt Ltd - Trivandrum, Kerala, India

20/11/2023 - 01/07/2024

- Worked on a research project to evaluate the therapeutic potential of *Garcinia gummi-gutta* extracts
- Assays conducted were:
 - Phytochemical Analysis: Conducted both qualitative and quantitative analysis
 - DPPH Assay, α -Amylase Inhibition Assay, MTT Assay (on MCF-7 cell line), Protein Denaturation Assay, Scratch Wound Assay (on L929 cell line), Acid Neutralizing Capacity Method, Agar Well Diffusion Method
- Cell Culture: Hands-on experience with MCF-7 and L929 cell lines

Validation Analyst Zifo Rnd Solutions - Chennai, Tamil Nadu, India

02/08/2021 - 20/10/2023

- Validated scientific software platforms including ELN, LIMS and laboratory instrument systems used within pharmaceutical R&D and manufacturing environments.
- Performed software validation activities in compliance with FDA 21 CFR Part 11, EU Annex 11 and GxP requirements.

Intern ICAR - Central Tuber Crops Research Institute - Trivandrum, Kerala, India

06/07/2018 - 13/07/2018

- Hands on training on 'Genomics and Molecular Biology Laboratory Technique' such as different streaking methods, PCR, Restriction digestion, plasmid isolation from *E. coli*, bacterial transformation, and DNA extraction from Cassava leaves, along with expertise in tissue culture.

PUBLICATIONS

Abhirami A. B., Parvathy Prasad. Evaluating The Therapeutic Potential Of *Garcinia Gummi-Gutta*. International Journal in Pharmaceutical Sciences. 2024Aug31;2(8):4058–82.

CONFERENCE PRESENTATIONS

- Poster Presentation, Imperial College London (2026)
CDK12/13 Inhibition Disrupts DNA Damage Repair Pathways in MYCN-Amplified Neuroblastoma
- Oral Presentation, BRIO 2021 National Online Conference
Bibliometric Study and Molecular Docking of Ferulic Acid with Alzheimer's Disease Causing Proteins

TECHNICAL SKILLS

Cell & Molecular Biology:

- Mammalian cell culture (2D & 3D) • Drug treatment studies • Cell viability assays • Spheroid culture • RNA extraction and cDNA synthesis • DNA extraction • RT-qPCR and RT-PCR • Alternative splicing analysis (exon skipping and intron retention) • Transient plasmid transfection • Primer design and validation • CRISPR-Cas9-mediated gene knockdown

Protein Biology

- Western blotting • SDS-PAGE • Protein quantification • ELISA

Microscopy & Imaging

- Immunofluorescence • Confocal microscopy • InCell imaging

Functional Assays

- Caspase-Glo 3/7 assay • HR Reporter assay • Flow cytometry

Data Analysis

- Experimental design and optimization • GraphPad Prism • ImageJ/Fiji • FlowJo • Cell Profiler • RNA-seq data interpretation • Microsoft Excel

Digital Systems

- ELN • LIMS

AWARDS & SCHOLARSHIPS

- Prime Minister's Scholarship Scheme (4 years)
- Kerala State Council for Science, Technology and Environment Research Grant
- NPTEL Topper – Applied Environmental Microbiology course